

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18
- **Key Features:**
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- **Missing Data:** 37 values in the Review Rating column.

3. Exploratory Data Analysis using Python

Data preparation and cleaning were performed using Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	1
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	0
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

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Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to `snake_case` for better readability and documentation.
- **Feature Engineering:** * Created an `age_group` column by binning customer ages.
 - Created a `purchase_frequency_days` column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected the Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Gender

	gender text 	revenue numeric 
1	Female	75191
2	Male	157890

- **Female:** \$75,191
- **Male:** \$157,890

2. High-Spending Discount Users

Identified 839 customers who used discounts but still spent above the average purchase amount.

	customer_id bigint 	purchase_amount bigint 
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:00

3. Top 5 Products by Rating

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

1. **Gloves:** 3.86
2. **Sandals:** 3.84
3. **Boots:** 3.82
4. **Hat:** 3.80
5. **Skirt:** 3.78

4. Shipping Type Comparison

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

- **Standard:** \$58.46
- **Express:** \$60.48

5. Subscribers vs. Non-Subscribers

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

- **Subscribed (Yes):** 1,053 customers | Avg. Spend: \$59.49 | Total Revenue: \$62,645.00
- **Unsubscribed (No):** 2,847 customers | Avg. Spend: \$59.87 | Total Revenue: \$170,436.00

6. Discount-Dependent Products

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

1. **Hat:** 50.00%
2. **Sneakers:** 49.66%
3. **Coat:** 49.07%
4. **Sweater:** 48.17%
5. **Pants:** 47.37%

7. Customer Segmentation

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

- **Loyal:** 3,116 customers
- **Returning:** 701 customers
- **New:** 83 customers

8. Top 3 Products per Category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

- **Accessories:** Jewelry (171), Sunglasses (161), Belt (161)
- **Clothing:** Blouse (171), Pants (171), Shirt (169)
- **Footwear:** Sandals (160), Shoes (150), Sneakers (145)
- **Outerwear:** Jacket (163), Coat (161)

9. Repeat Buyers & Subscriptions

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

- **No Subscription:** 2,518 repeat buyers
- **Subscribed:** 958 repeat buyers

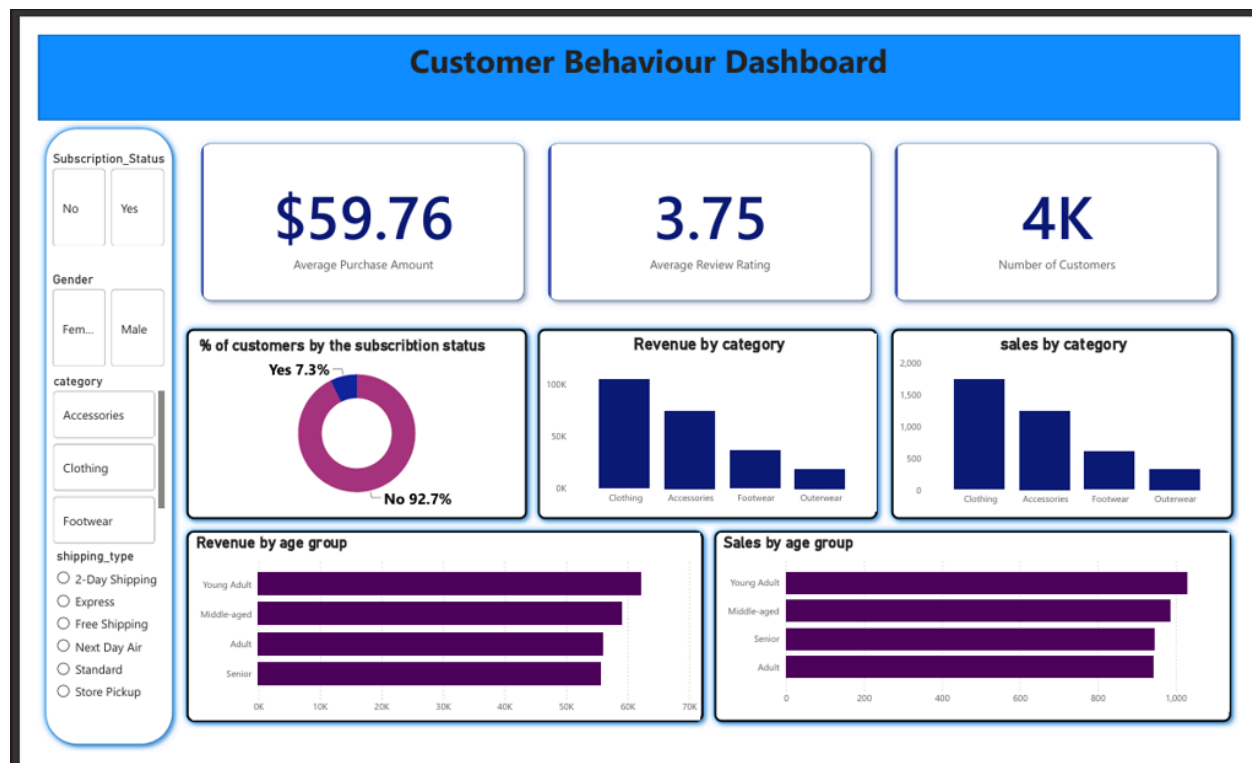
10. Revenue by Age Group

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

- **Young Adult:** \$62,143
- **Middle-aged:** \$59,197
- **Adult:** \$55,978
- **Senior:** \$55,763

5. Dashboard in Power BI

An interactive dashboard was built in Power BI to present these insights visually.



Core Metrics Tracked:

- **Average Purchase Amount:** \$59.76
- **Average Review Rating:** 3.75
- **Total Number of Customers:** 4K (Rounded from 3.9K)

6. Business Recommendations

- **Boost Subscriptions:** Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs:** Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy:** Balance sales boosts with margin control.
- **Product Positioning:** Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing:** Focus efforts on high-revenue age groups and express-shipping users.